

Part A

Series Circuit (20Ω , 30Ω , 50Ω)

Total Resistance

$$20 + 30 + 50 = 100\Omega$$

Total Current

$$I = \frac{V}{R} = \frac{125}{100} = 1.25 \text{ A}$$

Current through Each Resistor

$$I_{20} = I_{30} = I_{50} = 1.25 \text{ A}$$

Voltage drop across each resistor ($V = IR$)

Resistor Voltage Drop

$$20\Omega \quad 1.25 \times 20 = 25 \text{ V}$$

$$30\Omega \quad 1.25 \times 30 = 37.5 \text{ V}$$

$$50\Omega \quad 1.25 \times 50 = 62.5 \text{ V}$$

$$25 + 37.5 + 62.5 = 125 \text{ V}$$

Power Dissipated ($P = V \times I$)

Resistor Power

$$20\Omega \quad 25 \times 1.25 = 31.25 \text{ W}$$

$$30\Omega \quad 37.5 \times 1.25 = 46.875 \text{ W}$$

$$50\Omega \quad 62.5 \times 1.25 = 78.125 \text{ W}$$

$$31.25 + 46.875 + 78.125 = 156.25 \text{ W}$$

Circuit B

Total Resistance

$$\frac{1}{R} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50} = \frac{5}{100} + \frac{1}{100} + \frac{2}{100} = \frac{8}{100}$$

$$\frac{100}{8} = 12.5 \Omega$$

Total Current

$$I_t = \frac{125}{12.5} = 10 \text{ A}$$

Current through each Resistor

Resistor	Current
20 Ω	$125/20 = 6.25 \text{ A}$
50 Ω	$125/50 = 2.5 \text{ A}$
100 Ω	$125/100 = 1.25 \text{ A}$

Voltage drop across each Resistor

$$V_{20} = V_{50} = V_{100} = 125 \text{ V}$$

Power Dissipated ($P = V \times I$)

Resistor	Power
20 Ω	$125 \times 6.25 = 781.25 \text{ W}$
50 Ω	$125 \times 2.5 = 312.5 \text{ W}$
100 Ω	$125 \times 1.25 = 156.25 \text{ W}$

$$781.25 + 312.5 + 156.25$$

$$1250 \text{ W}$$